

The Type-II Projection-Lattice Conjectures of arXiv:2006.08959

Fully answered by arXiv:2010.01627 and arXiv:2206.00875

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Status: literature already answered. No new theorem is claimed. Ayupov–Kudaybergenov settle the finite type-II₁ case, and Mori settles the remaining properly infinite type-II_∞ case. Together they prove the source’s stronger Conjecture 5.1 and consequently its weaker Conjecture 5.2.

1 Source conjectures

For a von Neumann algebra M , let $LS(M)$ be its algebra of locally measurable operators and $\mathcal{P}(M)$ its projection lattice. Mori’s source paper, arXiv:2006.08959, proves the desired structural theorem outside type II and ends with the two conjectures reproduced below.

results to a non-continuous ring isomorphism Ψ_n from $LS(N)$ onto $LS(M)$, $n \geq 1$. By Lemma 2.1 we may verify the statement for each Ψ_n , which suffices to complete the proof. $\square$

Corollary 4.3. *Let M, N be von Neumann algebras of type I_∞ or III. Suppose that $\Phi: \mathcal{P}(M) \rightarrow \mathcal{P}(N)$ is a lattice isomorphism. Then there exist a real $*$ -isomorphism $\psi: M \rightarrow N$ and an invertible element $y \in LS(N)$ such that $\Phi(p) = l(y\psi(p))$, $p \in \mathcal{P}(M)$.*

5. QUESTIONS

The author skeptically conjectures that the same as Theorem B holds for type II von Neumann algebras:

Conjecture 5.1. *Let M and N be von Neumann algebras of type II. Suppose that $\Psi: LS(M) \rightarrow LS(N)$ is a ring isomorphism. Then there exist an invertible operator $y \in LS(N)$ and a real $*$ -isomorphism $\psi: M \rightarrow N$ such that $\Psi(x) = y\psi(x)y^{-1}$ for any $x \in LS(M)$.*

Not much is known about the structure of the algebra $LS(M)$ for a type II (in particular, II₁) von Neumann algebra M . The author does not know whether or not such a Ψ is automatically real-linear even in the case M and N are (say, approximately finite dimensional) II₁ factors. Note that $LS(M)$ cannot have a Banach algebra structure because of the fact that an element of $LS(M)$ can have an empty or dense spectral set. Hence it seems to be difficult to make use of *automatic continuity* results on algebra isomorphisms as in [2]. However, the author suspects that at least the following weaker statement holds:

Conjecture 5.2. *Let M and N be von Neumann algebras of type II. If $\mathcal{P}(M)$ and $\mathcal{P}(N)$ are lattice isomorphic, or equivalently, if $LS(M)$ and $LS(N)$ are ring isomorphic, then M and N are real $*$ -isomorphic (or equivalently, M and N are Jordan $*$ -isomorphic).*

In another direction, we compare Theorem A with von Neumann’s theory of complemented modular lattices and regular rings. Von Neumann axiomatized projection lattices of type II₁ von Neumann algebras, and completed the amazing theory on the correspondence between the vast classes of complemented modular lattices and regular rings. Let us briefly recall this theory in [18] Part II].

Definition 5.3. A lattice L with smallest element 0 and least element 1 is called

Figure 1: Conjectures 5.1 and 5.2 on source PDF page 19 (printed page 9) of arXiv:2006.08959v1.

The stronger statement is:

$$\begin{aligned} M, N \text{ of type II, } \quad \Psi : \text{LS}(M) \longrightarrow \text{LS}(N) \text{ a ring isomorphism} \\ \implies \quad \Psi(x) = y\psi(x)y^{-1}, \end{aligned}$$

where $y \in \text{LS}(N)$ is invertible and $\psi : M \rightarrow N$ is a real $*$ -isomorphism extending to the locally measurable algebras. Conjecture 5.2 asks only whether a lattice isomorphism $\mathcal{P}(M) \cong \mathcal{P}(N)$ forces M, N to be real $*$ -isomorphic, equivalently Jordan $*$ -isomorphic.

2 The finite type-II₁ answer

Ayupov and Kudaybergenov, arXiv:2010.01627, explicitly say that their main result confirms Conjecture 5.1 of the source. Their Theorem 1.4 states that for type-II₁ algebras M, N , every ring isomorphism $\Phi : S(M) \rightarrow S(N)$ has the form

$$\Phi(x) = a\Psi(x)a^{-1}$$

for an invertible $a \in S(N)$ and a real $*$ -isomorphism $\Psi : M \rightarrow N$ extending to $S(M) \rightarrow S(N)$. Since a finite von Neumann algebra satisfies $\text{LS}(M) = S(M)$, this is exactly the type-II₁ part of Conjecture 5.1. Corollary 1.5 gives the projection-lattice equivalence requested in Conjecture 5.2.

In Section 2 we give definitions of various kinds of isomorphisms between $*$ -algebras and also some preliminaries from the theory of measurable operators affiliated with von Neumann algebras.

In order to prove the main result of the present paper, in Sections 3 and 4 we show automatic real-linearity and automatic continuity of ring isomorphisms between algebras of measurable operators affiliated with von Neumann algebras of type II₁. Namely, we prove the following two theorems.

Theorem 1.2. *Let $\mathcal{M}$ and $\mathcal{N}$ be type II₁ von Neumann algebras. Then any ring isomorphism from $S(\mathcal{M})$ onto $S(\mathcal{N})$ is a real algebra isomorphism.*

Theorem 1.3. *Let $\mathcal{M}$ and $\mathcal{N}$ be type II₁ von Neumann algebras. Then any ring isomorphism from $S(\mathcal{M})$ onto $S(\mathcal{N})$ is continuous in the local measure topology.*

In Section 5 the following main result confirms the Conjecture 5.1 in [13] and answers the above Question 1.1 for the type II₁ case.

Theorem 1.4. *Let $\mathcal{M}$ and $\mathcal{N}$ be von Neumann algebras of type II₁. Suppose that $\Phi : S(\mathcal{M}) \rightarrow S(\mathcal{N})$ is a ring isomorphism. Then there exist an invertible element $a \in S(\mathcal{N})$ and a real $*$ -isomorphism $\Psi : \mathcal{M} \rightarrow \mathcal{N}$ (which extends to a real $*$ -isomorphism from $S(\mathcal{M})$ onto $S(\mathcal{N})$) such that $\Phi(x) = a\Psi(x)a^{-1}$ for all $x \in S(\mathcal{M})$.*

Corollary 1.5. *Let $\mathcal{M}$ and $\mathcal{N}$ be von Neumann algebras of type II₁. The projection lattices $P(\mathcal{M})$ and $P(\mathcal{N})$ are lattice isomorphic, if and only if the von Neumann algebras $\mathcal{M}$ and $\mathcal{N}$ are real $*$ -isomorphic (or equivalently, $\mathcal{M}$ and $\mathcal{N}$ are Jordan $*$ -isomorphic).*

2. PRELIMINARIES

Figure 2: Theorems 1.2–1.4 and Corollary 1.5 on PDF page 2 of arXiv:2010.01627v2.

3 The type-II_∞ answer

Mori's later paper arXiv:2206.00875 identifies II_∞ as the last blank case. Theorem 1.2 proves that if M, N are of type II_∞, every ring isomorphism $\Psi : \text{LS}(M) \rightarrow \text{LS}(N)$ is

$$\Psi(x) = y\psi(x)y^{-1}$$

with $y \in \text{LS}(N)$ invertible and $\psi : M \rightarrow N$ a real $*$ -isomorphism extending to $\text{LS}(M) \rightarrow \text{LS}(N)$.

ring isomorphisms between their centers [11] Proposition 4.2] (see also [11]). The center of $LS(M)$ is in fact identified with the algebra of all complex-valued measurable functions on a measure space (modulo the almost everywhere equivalence relation). Getting a further concrete description of the general form of the latter ring isomorphisms seems to be out of the reach of the theory of operator algebras (see the remark right after [11] Proposition 4.2]).

The author proved that $\Psi: LS(M) \rightarrow LS(N)$ is similar to a real $*$ -isomorphism whenever M and N are type I_∞ or III von Neumann algebras [11] Theorem B]. See [11] for a more detailed overview of the above results and the history of related topics. In the same paper, the author left the case of type II as an open question, conjecturing that every ring isomorphism $\Psi: LS(M) \rightarrow LS(N)$ is similar to a real $*$ -isomorphism when M and N are of type II [11] Conjecture 5.1]. This conjecture was soon confirmed by Ayupov and Kudaybergenov in the case of type II_1 von Neumann algebras [2] Theorem 1.4].

The purpose of this article is to fill in the still blank case of type II_∞ . Namely, the main theorem is as follows.

Theorem 1.2. *Let M, N be von Neumann algebras of type II_∞ . If $\Psi: LS(M) \rightarrow LS(N)$ is a ring isomorphism, then there exist a real $*$ -isomorphism $\psi: M \rightarrow N$ (which extends to a real $*$ -isomorphism from $LS(M)$ onto $LS(N)$) and an invertible element $y \in LS(N)$ such that $\Psi(x) = y\psi(x)y^{-1}$, $x \in LS(M)$.*

This together with [11] Theorem B] and [2] Theorem 1.4] verifies

Theorem 1.3. *Let M, N be von Neumann algebras without type I_{fin} direct summands. If $\Psi: LS(M) \rightarrow LS(N)$ is a ring isomorphism, then there exist a real*

Figure 3: Theorem 1.2 and the combined theorem on PDF page 2 of arXiv:2206.00875v2.

The same page explicitly combines this theorem with Ayupov–Kudaybergenov’s Theorem 1.4 and the source’s Theorem B. Its Theorem 1.3 gives the inner-twisted real- $*$ form for all von Neumann algebras without finite type-I direct summands, and Theorem 1.4 gives the corresponding classification of projection-lattice isomorphisms. In particular both source conjectures hold for every type-II algebra, including arbitrary central mixtures of finite and properly infinite type-II summands.

4 Conclusion and verification

The later results match the source quantifiers and conclusion exactly:

Source scope	Later theorem	Status
type II_1	arXiv:2010.01627, Thm. 1.4	answered
type II_∞	arXiv:2206.00875, Thm. 1.2	answered
all type II	arXiv:2206.00875, Thms. 1.3–1.4	answered

The source and both supporting PDFs were downloaded from arXiv. The displayed source passage and theorem statements were checked directly against those PDFs, not inferred from abstracts alone. A bounded search of the run indexes found no prior packet for arXiv:2006.08959. The source conjectures therefore require no new proof attempt and are classified `literature_already_answered`.

References

- [1] M. Mori, *Lattice isomorphisms between projection lattices of von Neumann algebras*, Forum Math. Sigma 8 (2020), e49; arXiv:2006.08959.
- [2] Sh. A. Ayupov and K. K. Kudaybergenov, *Ring isomorphisms of Murray–von Neumann algebras*, J. Funct. Anal. 280 (2021), 108891; arXiv:2010.01627.
- [3] M. Mori, *Ring isomorphisms of type II_∞ locally measurable operator algebras*, Bull. Lond. Math. Soc. 55 (2023), 2525–2538; arXiv:2206.00875.